

# Global Airbnb Performance Dashboard

## Data Analysis & Findings Report

Prepared By:

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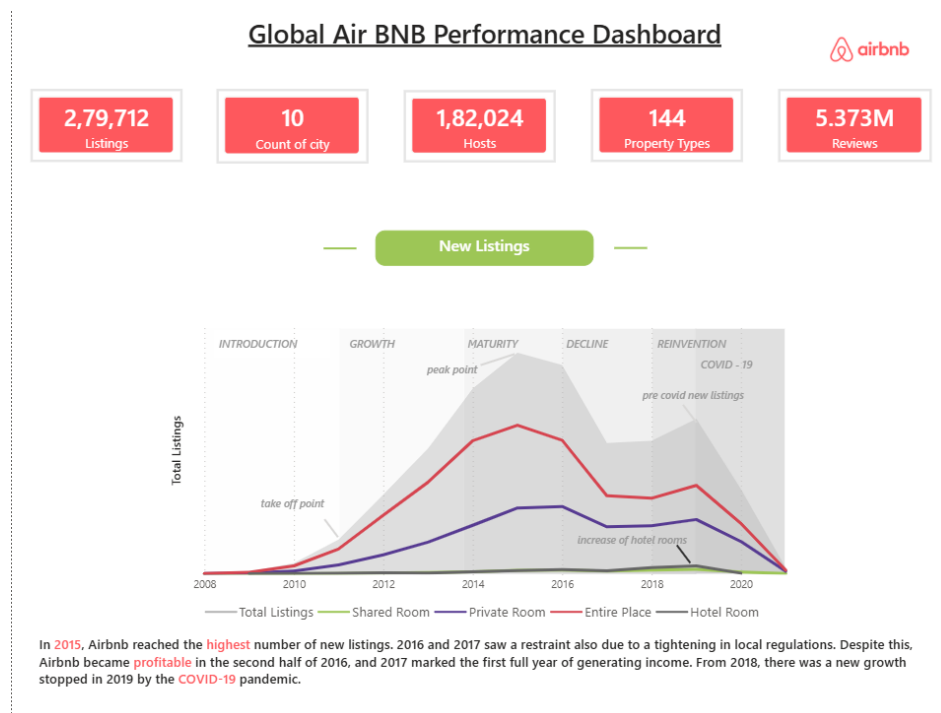
### 1. Executive Summary

The Global Airbnb Performance Dashboard was developed using Power BI to analyze Airbnb listings, host activity, customer engagement, and market trends across multiple cities worldwide.

The objective of this project was to transform raw Airbnb data into a centralized analytical solution that supports strategic business decisions related to platform growth, customer behavior, host trust, and seasonal demand patterns.

Through interactive dashboards and KPI-driven insights, the analysis reveals how Airbnb evolved over time, which property categories dominate the marketplace, how customers engage with listings, and how seasonality impacts demand across regions.

The dashboard enables stakeholders to identify operational risks, evaluate market opportunities, and improve pricing and expansion strategies using data-driven insights.



## 2. Project Objectives

The primary objectives of this analysis were:

- Track Airbnb listing growth over time
  - Analyze customer engagement through reviews
  - Understand host verification and trust patterns
  - Identify seasonal demand fluctuations
  - Compare performance across room types and cities
  - Provide actionable insights for business decision-making
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## 3. Dataset Overview

### Dataset Source

#### Airbnb Listings & Reviews – Maven Analytics

### Data Included

The dataset contains:

- Airbnb listing information
- Host details and verification data
- Customer reviews
- Property and room types
- Geographic information
- Seasonal and trend-related data

### Tools & Technologies Used

| Tool        | Purpose                             |
|-------------|-------------------------------------|
| Power BI    | Dashboard creation & visualization  |
| DAX         | KPI calculations and business logic |
| Power Query | Data transformation and cleaning    |

| Tool          | Purpose                 |
|---------------|-------------------------|
| Data Modeling | Relationship management |
| Excel / CSV   | Raw data storage        |

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#### 4. Data Cleaning & Transformation

Before analysis, the dataset underwent several preprocessing steps using Power Query and Power BI.

##### Data Preparation Activities

- Removed duplicate records
- Handled missing and null values
- Standardized date and categorical fields
- Created calculated measures using DAX
- Optimized relationships between tables
- Structured data model for efficient analysis

##### Key DAX Measures Created

- Total Listings
- Total Hosts
- Total Reviews
- Property Type Count
- Monthly Listing Growth
- Review Distribution Metrics

These transformations improved analytical accuracy and dashboard performance.

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#### 5. Key Performance Indicators (KPIs)

The dashboard includes several KPIs to monitor Airbnb platform performance.

##### Major KPIs

| KPI            | Description                                 |
|----------------|---------------------------------------------|
| Total Listings | Total number of Airbnb properties available |
| Total Hosts    | Total registered hosts                      |
| Total Reviews  | Total customer reviews across listings      |
| Property Types | Number of unique property categories        |
| Cities Covered | Number of cities included in analysis       |

These KPIs provide a high-level overview of Airbnb's operational scale and marketplace activity.

## 6. Dashboard Analysis & Findings

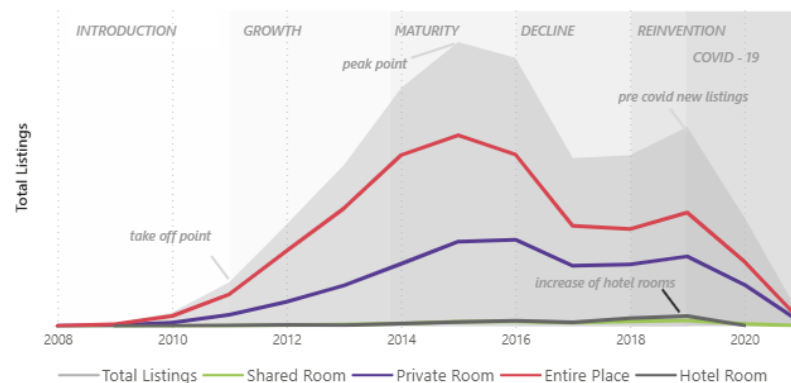
### 6.1 Listing Growth Analysis

#### Objective

To analyze how Airbnb listings evolved over time.

#### Findings

- Airbnb experienced rapid listing growth during the early expansion years.



- The platform reached peak growth around 2015.
- Growth slowed significantly during the COVID-19 period due to travel restrictions and reduced tourism demand.
- Recovery trends indicate resilience in the Airbnb marketplace after market disruptions.

## Business Insight

Understanding listing growth trends helps stakeholders evaluate market maturity, expansion opportunities, and operational scalability.

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## 6.2 Property Type & Room Type Analysis

### Objective

To identify the dominant property categories on the platform.

### Findings

- Entire Place listings dominate Airbnb's marketplace.



- Shared rooms and hotel rooms represent a much smaller market share.
- Customers show a strong preference for privacy and independent accommodations.

## Business Insight

Airbnb can prioritize marketing and pricing strategies around high-demand property categories to maximize platform performance.

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## 6.3 Customer Engagement & Review Behavior

### Objective

To evaluate customer interaction and engagement patterns.

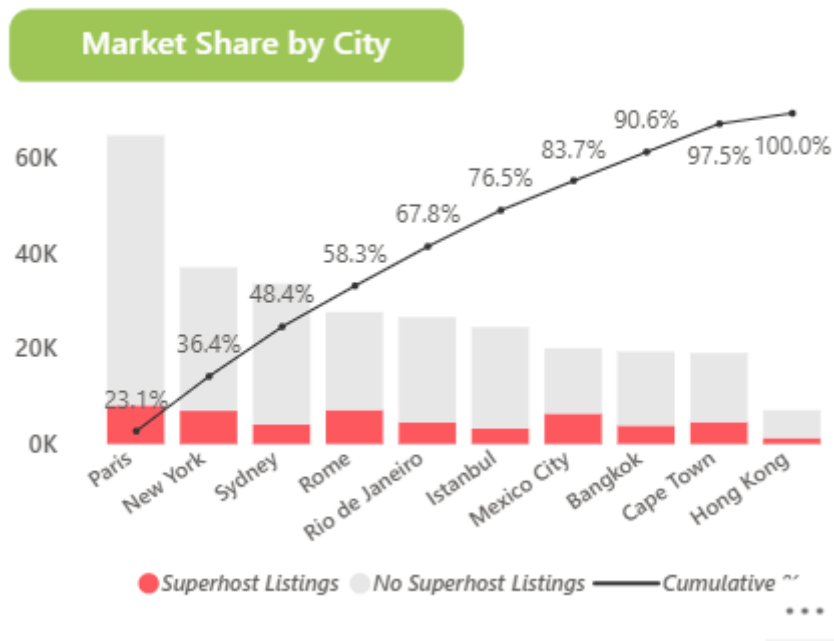
### Findings

- Review frequency indicates active customer participation across the platform.
- Certain listings receive significantly higher review counts, suggesting stronger customer satisfaction or popularity.

- Repeat reviewer behavior suggests the presence of highly engaged users and frequent travelers.

### Business Insight

Customer engagement metrics help identify top-performing listings and improve customer retention strategies.



## 6.4 Host Trust & Verification Analysis

### Objective

To assess trust and safety indicators among hosts.

### Findings

- Verified hosts contribute significantly to customer trust.
- Listings associated with verified hosts generally show stronger engagement patterns.
- Some regions exhibit lower verification rates, which may represent operational or safety risks.

### Business Insight

Trust and verification analytics help Airbnb improve safety policies and strengthen customer confidence in the platform.

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## 6.5 Seasonality & Demand Trends

### Objective

To identify seasonal demand fluctuations across Airbnb markets.

### Findings

- Demand varies significantly across months and geographic regions.
- Peak seasonal demand aligns with tourism and holiday periods.
- Off-season months demonstrate lower review and listing activity.

### Business Insight

Seasonality insights can support dynamic pricing strategies, resource planning, and promotional campaigns.

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## 7. Business Impact

The dashboard delivers significant business value by enabling stakeholders to make informed strategic decisions.

### Key Business Benefits

- Supports market expansion analysis
- Enhances operational planning
- Improves pricing optimization strategies
- Strengthens trust and safety monitoring
- Helps identify customer behavior trends
- Enables data-driven decision-making

The dashboard converts large volumes of raw Airbnb data into meaningful insights that improve analytical visibility across the business.

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## 8. Analytical Questions Answered

This project successfully addresses several important business questions:

- How has Airbnb grown over time?
- Which room types dominate the platform?

- What are the customer engagement trends?
  - How important is host verification?
  - How does demand change seasonally?
  - Which markets show stronger activity levels?
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## **9. Challenges Faced**

During the development process, several challenges were encountered:

- Managing large datasets efficiently
- Cleaning inconsistent and incomplete records
- Optimizing Power BI model relationships
- Building meaningful DAX measures
- Designing intuitive and interactive visualizations

These challenges were resolved through structured data modeling and transformation techniques.

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## **10. Future Improvements**

Potential future enhancements for this project include:

- Predictive analytics for booking trends
- Real-time Airbnb API integration
- City-level drill-through dashboards
- Demand forecasting models
- Advanced customer segmentation analysis
- Machine learning-based pricing recommendations

These improvements could further enhance the dashboard's analytical capabilities.

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## **11. Conclusion**

The Global Airbnb Performance Dashboard successfully transforms Airbnb marketplace data into an interactive business intelligence solution.



The analysis highlights critical insights related to listing growth, customer engagement, host verification, property trends, and seasonal demand behavior. Through effective use of Power BI, DAX, and data modeling techniques, the project demonstrates how business intelligence tools can support strategic planning and operational decision-making.

This dashboard provides stakeholders with a centralized, data-driven approach to understanding Airbnb's global performance and identifying future growth opportunities.

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## **12. Author**

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Power BI | Data Analytics | Business Intelligence